

Progress Form

Group no. 14694.

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Project Title Masar: AI-Powered Pilgrim Tracking and Safety System

AI Domain for the Senior project: Deep Learning for Time-Series Anomaly Detection

Modified Abstract:

Mass gatherings such as the Hajj pilgrimage create significant challenges in maintaining group cohesion and ensuring pilgrim safety due to dense crowds and continuous movement between constrained locations. Traditional campaign supervision methods rely heavily on manual coordination and basic communication tools, which do not provide automated monitoring or early detection of emerging risks. This paper presents Masar, a lightweight real-time monitoring system designed to enhance campaign-level supervision through automated GPS-based tracking and behavioral analysis.

The system enables background location updates once a device is registered, eliminating the need for continuous user interaction. To ensure practical deployment during long pilgrimage activities, Masar incorporates adaptive sampling to reduce battery consumption and local data buffering to handle temporary network interruptions. Collected GPS streams are transformed into sequential movement features such as displacement, speed variation, and directional change. A Long Short-Term Memory (LSTM) model analyzes these temporal patterns to detect abnormal mobility behavior.

To improve reliability under noisy real-world conditions, the system integrates a persistence-based anomaly filter that triggers alerts only when abnormal predictions remain consistent across consecutive time windows. Additionally, Masar introduces time-based risk escalation, where alert severity increases progressively if irregular behavior persists. Group awareness mechanisms further evaluate each pilgrim relative to the group's centroid to detect gradual separation before critical divergence occurs.

The end-to-end architecture combines live data ingestion, sequential inference, and a web-based supervisory dashboard that visualizes multi-level risk indicators in real time. Experimental evaluation demonstrates that the proposed approach provides stable and interpretable abnormal behavior detection while maintaining a lightweight and deployable design. Masar offers a scalable framework for intelligent group monitoring in Hajj campaigns and similar large-scale events requiring structured mobility supervision.

Research Objectives: (with progress % bar)

Please write your research objective, how much you have accomplished in SP1 and the new objectives you intend to work on in SP2. This should include all Project objectives that have been written in the SP1 report and any more objectives you wish to add in SP2.

Research Objectives for SP1		Progress %
Objective	Progress Bar	Progress %
System architecture design		100%
Dataset preparation & preprocessing		100%
Model training and evaluation		100%
Real-time Firebase pipeline		100%
Supervisor dashboard implementation		95%
MVP testing and validation		90%

Research Objectives for SP2		Progress %
Objective	Progress Bar	Progress %
Automatic background tracking		10%
Persistence-based anomaly filtering		5%
Time-based risk escalation		5%
Group-awareness metrics		5%
Adaptive sampling optimization		0%
Offline buffering support		0%
Extended evaluation		0%

The Template for the final report:

Please write one (or two maximum) conferences/journals (or Patent draft) that you can target to submit your paper by the end of the semester. Your chosen conference or journal (or Patent draft) must follow the following requirements:

- 1- The chosen paper must follow the research standards of the university of Jeddah.
- 2- The length of the 2-column paper (or Patent draft) template should be at least ten pages without references and appendixes.

3- The deadline for submission to the chosen conference should be the 16th week of the Sst semester at the earliest.

4- The journal should not be predatory, Please check in the following link
<https://beallslist.net/standalone-journals/#update>

5- The affiliation of the university must be written properly following the **University of Jeddah, Author affiliation and acknowledgements guidelines.**

In the following link you can find all call for papers.
<http://www.wikicfp.com/cfp/>

Your Template will follow the **Conference** / Journal:

IEEE International Smart Cities Conference (ISC2) - we will follow the official IEEE 2-column conference paper template (IEEEtran) used in this venue. Template

link: <https://www.ieee.org/conferences/publishing/templates.html>

